

POLARIS DC-1 — SITE SURVEY, GEOTECHNICAL AND CONSTRAINTS REPORT

Document Ref: PDC-SS-001 Rev B
Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
Prepared for: Polaris Digital Infrastructure Oy (the Employer)
Prepared by: Maaperä Geotek Oy, Consulting Geotechnical and Geoenvironmental Engineers, Helsinki
Report date: 21 November 2025
Status: Employer's document — issued for design and tender purposes

Synthetic demonstration data. This report and all findings, parties, borehole records and analytical results within it are fictional and were generated to exercise the White Hat IQ platform. No real site, company or individual is described.

1. Scope, Investigation Dates and Standards

1.1 Scope of this report

Maaperä Geotek Oy was commissioned by Polaris Digital Infrastructure Oy to carry out a desk study, intrusive ground investigation and geotechnical appraisal of the proposed data campus site at Aviapolis, Vantaa, to inform the Employer's Requirements and the Contractor's detailed design. The scope covers: ground conditions and stratigraphy; groundwater; rock quality and excavatability; preliminary foundation recommendations; and a register of physical and regulatory site constraints. This report does not address contamination beyond the observations noted in Section 7, drainage design, or any contractual or commercial matter — those are addressed elsewhere in the project document set.

1.2 Investigation programme and revision history

Item	Detail
Desk study	September 2025
Intrusive investigation (Phase 1, BH-01 to BH-10)	6–17 October 2025
Intrusive investigation (Phase 2, BH-11 to BH-14, targeted infill)	27–31 October 2025
Laboratory testing	October–November 2025
Rev A issued	4 November 2025 (superseded)
Rev B issued (this report)	21 November 2025 — incorporates Phase 2 infill boreholes and sulphide sample results at BH-12

1.3 Standards and guidance applied

- SFS-EN 1997-1 and SFS-EN 1997-2 (Eurocode 7), Finnish National Annexes
- SFS-EN ISO 14688-1/-2 — identification, description and classification of soil
- SFS-EN ISO 22475-1 — sampling and groundwater measurement

- RIL 121-2004 and RIL 263-2014 — Finnish Geotechnical Society guidance on ground investigation and excavation/support
- SFS 4812 — vibration from blasting and piling, guide values near structures
- Land Use and Building Act (132/1999) and associated Government Decrees

2. Desk Study and Site History

The site occupies Plot 92-52-7-2 in the Aviapolis business district of Vantaa, approximately 4.8 ha, and is bounded to the north by a 110 kV overhead transmission line, to the north-east by residential development, to the south-west by municipal services infrastructure, and to the west by Tikkurilantie and an operating logistics facility. The plot falls approximately 4 m from north to south.

Review of historical aerial photography (1962–2019) and municipal planning records indicates the site was under coniferous forest cover throughout this period, with no record of quarrying, landfilling, industrial use or structures. The site was clear-felled in 2019 in preparation for development and has remained undeveloped scrubland since. No environmental permits, contaminated-land register entries or historical industrial activity were identified for the plot in the Finnish Environment Institute (SYKE) or Vantaa municipal records reviewed. On this basis the site was assessed as low risk for contamination prior to intrusive works; this was subsequently qualified by the localised observations at Section 7.

Regional geology (Geological Survey of Finland, GTK bedrock and Quaternary maps) shows the area underlain by Svecofennian granitic gneiss with a discontinuous cover of glacial till, thickening toward the west and thinning toward the east across the wider Aviapolis area — consistent with the pattern found at this site (Section 3).

3. Ground Investigation Summary

Fourteen boreholes (BH-01 to BH-14) were advanced across the site using rotary percussion drilling with Standard Penetration Testing (SPT) at 1.0–1.5 m intervals, supplemented by rotary core drilling into rock at boreholes where rockhead was proven. Borehole positions were set out on a local site grid (origin at the south-west site corner, Easting/Northing in metres) to give reasonable coverage of the plot while concentrating coverage in the eastern third, where shallower rock was anticipated from the desk study.

3.1 Borehole schedule

ID	Grid location (E, N)	Ground level (mAMSL)	Depth to rock (m)	Till description	SPT N-value range	Groundwater depth (m)	Remarks
BH-01	E030, N145	21.6	9.2	Loose brown reworked fill 0.0–1.7 m over firm, becoming stiff, sandy gravelly till	8–15 (fill); 20–42 (till)	3.1	Made ground encountered; NW corner — see Section 7b
BH-02	E075, N150	21.8	8.6	Firm becoming stiff sandy gravelly till with scattered cobbles	18–40, increasing with depth	3.4	Edge of made-ground area; fill not encountered here

BH-03	E150, N152	21.3	7.4	Stiff sandy gravelly till, occasional cobbles	24–46	3.0	—
BH-04	E215, N148	20.9	6.1	Stiff gravelly till, cobbles frequent near base	28–50	2.8	110 kV line crosses boundary approx. 40 m north
BH-05	E272, N135	20.2	4.6	Very stiff sandy till, frequent cobbles and boulders near rockhead	35–refusal (N>50/300 mm)	2.7	East zone, shallow rock — see Section 5
BH-06	E045, N092	20.4	8.9	Firm to stiff sandy gravelly till	16–38	3.6	—
BH-07	E140, N096	19.9	Not proven (boulder refusal at 7.8)	Stiff sandy till; dense boulder horizon 6.5–7.8 m	26–48, refusal on obstruction	3.3	Boulder-rich horizon, rock level not proven — see Section 7a
BH-08	E192, N090	19.6	Not proven (boulder refusal at 6.4)	Stiff sandy gravelly till; boulders from 6.0 m	30–49, refusal on obstruction	3.2	Boulder-rich horizon, rock level not proven — see Section 7a
BH-09	E243, N086	19.3	5.0 (rockhead proven by core)	Stiff till with scattered boulders 4.0–5.0 m above proven rockhead	32–50	2.9	Boulders above proven rock — see Section 7a
BH-10	E062, N042	18.4	8.7	Thin topsoil/organic horizon 0.0–0.6 m over firm sandy gravelly till	6–34	2.6	Near SW water/foul connection point
BH-11	E132, N037	18.2	7.6	Stiff sandy gravelly till	22–45	3.0	—
BH-12	E203, N031	18.0	5.8	Stiff grey till; dark grey/black sulphide-stained horizon 3.2–4.6 m, faint sulphurous odour noted on opening	27–47	2.8	Elevated sulphide suspected, sample sent for analysis — see Section 7c
BH-13	E262, N038	18.6	4.4	Very stiff sandy till, frequent boulders and cobbles	36–refusal (N>50/280 mm)	2.9	East zone, shallow rock, within anticipated blasting area
BH-14	E284, N103	19.5	4.0	Very stiff sandy till, thin cover directly over rockhead	40–refusal (N>50/150 mm)	3.1	Shallowest proven rock on site; east boundary

Ground levels fall from approximately 21.8 mAMSL at the north of the plot to 18.0 mAMSL at the south, consistent with the recorded overall fall of approximately 4 m. Depth to rock ranges from 4.0 m to 9.2 m across the fourteen boreholes and is not uniform: the shallowest values (4.0–5.0 m) are consistently found in the eastern third of the plot (BH-05, BH-09, BH-13, BH-14), with depths generally increasing toward the west and north-west (up to 9.2 m at BH-01). This pattern is consistent with the regional Quaternary geology reviewed at desk study stage.

4. Groundwater

Groundwater was recorded in all boreholes as a perched water table within the till, at depths ranging from 2.6 m to 3.9 m below existing ground level at the time of drilling (October 2025). No continuous regional water table

was identified within the depths investigated; the till is of variable permeability and the recorded levels are interpreted as perched water above less permeable till horizons rather than a single coherent aquifer.

Standpipe piezometers installed in BH-02, BH-06, BH-10 and BH-12 will allow seasonal monitoring; based on regional experience for similar till sites in the Helsinki metropolitan area, seasonal variation of 0.5–1.0 m should be allowed for, with the highest levels typically occurring during spring snowmelt (April) and after prolonged autumn rainfall.

Dewatering implications: excavations for foundations, service trenches and the substation/generator compound basements (where below approximately 2.5 m) should anticipate perched groundwater ingress and allow for sump pumping or well-pointing as a minimum. Excavations extending into the boulder-rich horizons noted at BH-07/BH-08/BH-09 (Section 7a) may experience locally higher and less predictable seepage where boulders create preferential flow paths. No artesian conditions were recorded.

5. Rock: Description, Rippability and Blasting Requirement

Rock proven across the site is granitic gneiss, consistent with the regional Svecofennian basement. Rock quality varies from fair to good in recovered core, with RQD (Rock Quality Designation) values ranging from approximately 25% (locally, in a fractured zone noted in the BH-09 core run between 5.0 and 6.2 m) to 75% (BH-14, sound and closely to widely jointed). Unconfined compressive strength testing on selected core samples returned values in the range 110–190 MPa, indicating strong to very strong rock.

Rippability assessment: based on the recorded strength, jointing and RQD, the rock across the site — and specifically within the eastern third of the plot where rockhead is shallowest and most consistently encountered (BH-04, BH-05, BH-09, BH-13, BH-14) — falls outside the range considered rippable by conventional excavator-mounted rock breakers or ripping equipment. Mechanical excavation is assessed as **not viable** for bulk excavation of this rock. **Controlled blasting is therefore required** to achieve founding levels and service trench depths across the eastern third of the site.

Vibration and notification: the site is bounded to the north-east by residential development and, at the shared access from Tikkurilantie, by an operating logistics facility. Blasting works must be designed and monitored in accordance with SFS 4812 guide values for peak particle velocity (PPV) at the nearest sensitive receptors, with pre-blast condition surveys of the nearest residential structures and the logistics facility before works commence. A blasting notification (louhintailmoitus) will be required to the Vantaa building control and environmental authorities, together with advance written notice to affected neighbours, consistent with the Environmental Protection Act. Blast design, scaled-distance calculations and a vibration monitoring regime should be developed at construction stage by a licensed blasting contractor.

6. Foundation Recommendations

6.1 General approach

- **Where rock is deeper than approximately 6 m** (broadly the central and western two-thirds of the plot): conventional pad and strip footings founded on undisturbed, inorganic till are expected to be viable for the data hall and ancillary structures, subject to removal of topsoil, organic horizons and any made ground. Where column loads are high (e.g. transformer plinths, generator bases) piled foundations bearing within the till or socketed into rock should be considered as an alternative.
- **Within the eastern third of the plot**, where rock is shallow (4.0–6.1 m and locally less once blasted to formation level), direct bearing on sound blasted rock, or short rock-socketed piles/piers, is recommended for heavy plant foundations.
- **Slab-on-grade areas:** subgrade preparation should remove topsoil and any organic or made-ground material, followed by proof-rolling and placement of a compacted granular capping layer. In the north-west area affected by made ground (Section 7b), removal and replacement of the fill, or a piled slab solution, will be required pending confirmation of fill extent and composition.
- Preliminary values below are for design guidance; final bearing values should be verified at formation level during construction and confirmed against Eurocode 7 (Design Approach in use) by the Contractor's geotechnical designer.

6.2 Preliminary allowable bearing values

Material	Condition	Preliminary allowable bearing pressure
Topsoil / organic horizon	Any	Not suitable for founding — remove
Made ground / reworked fill (NW area)	Uncharacterised	Not suitable for direct founding — remove, or pile through
Firm to stiff sandy gravelly till	Founded ≥ 1.5 m below stripped level, away from boulder horizons	150 kPa
Stiff to very stiff till	Founded ≥ 3.0 m below stripped level	250 kPa
Very stiff till near rockhead	Founded immediately above proven rockhead	300–350 kPa
Sound granitic gneiss (unweathered, RQD >50%)	Direct bearing on prepared/blasted surface	2,500 kPa
Fractured/low-RQD gneiss (locally, e.g. near BH-09 5.0–6.2 m)	Direct bearing	To be assessed locally; dental concrete or deeper founding may be required

7. Residual Geotechnical Risks

Three matters could not be fully resolved within the scope of this investigation and are recorded here as residual risk for the Employer's and Contractor's attention rather than as investigation findings on which design can be finalised without further work.

7a. Boulder-rich horizon between BH-07 and BH-09 — rock level not proven

Both BH-07 and BH-08 met refusal on what is interpreted as boulder obstructions (at 7.8 m and 6.4 m respectively) without reaching proven rockhead; BH-09, approximately 55 m to the east, proved rock at 5.0 m but recorded scattered boulders in the till immediately above it. The investigation was not able to establish whether BH-07 and BH-08 are within

a discrete boulder nest or whether this reflects a broader boulder-rich transitional zone between the till and rockhead across this part of the site. **This is flagged as a risk to piling and bulk excavation productivity** in the affected area: boulders of this size can deflect or refuse driven or bored piles, increase blasting/breaking effort, and make excavation rates difficult to forecast reliably from the current borehole spacing. Additional targeted investigation (further boreholes and/or a geophysical survey, e.g. seismic refraction) between BH-07 and BH-09 is recommended before finalising the piling or bulk excavation methodology in this zone.

7b. Unrecorded made ground in the north-west corner — extent and archaeological potential unconfirmed

BH-01 encountered approximately 1.7 m of loose, reworked fill material directly beneath the ground surface, not present at the neighbouring BH-02 some 45 m to the south-east. Review of the available historical aerial imagery did not identify a clear source or date for this material, and no record of authorised filling on this plot was found in the municipal records reviewed. Based on the topography and the single confirmed borehole, the affected area is estimated at approximately 900 m² in the north-west corner of the plot, but **neither the full extent nor the depth of this material has been proven** by the current borehole spacing. Because the fill's origin and date cannot be confirmed from desk records, **the possibility that this area conceals earlier ground disturbance of archaeological interest cannot be excluded**, and it is considered likely that the Finnish Heritage Agency (Museovirasto / National Board of Antiquities) would require a watching brief or a preliminary archaeological assessment during topsoil strip and fill removal in this area. This should be factored into the enabling-works programme and permit strategy for this part of the site.

7c. Localised elevated sulphide indication near BH-12 — waste classification unconfirmed

A dark grey to black, sulphide-stained horizon was logged in BH-12 between 3.2 m and 4.6 m depth, with a faint sulphurous odour noted on opening the sample tubes. A sample from this horizon (SPH-12/3) has been submitted for total sulphur and acid-generation potential testing; results were not available at the time of issue of this report. Sulphide-bearing till and sulphide clay/soil (locally referred to as "mustaliuske"-associated or "happamat sulfaattimaat" conditions in Finnish practice, though the specific mineralogy has not yet been confirmed here) can, on exposure and oxidation, generate acidic leachate and may require classification as a different waste category for off-site disposal than the surrounding inorganic till. **This is flagged as a waste-classification and handling risk** for excavated material from this localised zone: confirmatory testing and, if confirmed, a specific handling and disposal routing plan should be in place before bulk excavation reaches this area. Pending confirmation, material from within approximately 15 m of BH-12 in this depth range should be segregated and stockpiled separately rather than mixed with general till arisings.

8. Site Constraints Register

Constraint	Location / extent	Impact
Bedrock at 4–9 m, shallowest in the east	Eastern third of the plot	Mechanical excavation not viable in the east; controlled blasting required (Section 5)
110 kV overhead transmission line	Crosses the northern site boundary	Exclusion zone for cranes and tall plant beneath and adjacent to the line; network operator's written permission required for any operation within its exclusion zone
Residential boundary, night-time noise limit 45 dB(A)	North-east boundary	Restricts night-time construction noise, including blasting timing and generator/plant testing, adjacent to this boundary
Aviation height limitation	Site-wide (Helsinki-Vantaa Airport approx. 3 km distant)	Constrains crane and permanent structure heights; obstacle lighting likely required on tall temporary and permanent plant
Single site access from Tikkurilantie, shared with an operating logistics neighbour	West boundary	Constrains delivery scheduling and traffic management; deliveries restricted to 06:00–22:00 on weekdays per the Employer's Requirements
Municipal water and foul connections	South-west corner	Fixes the practical connection point for site drainage and water supply design; BH-10 (nearest borehole) recorded a thin organic surface horizon in this area
Perched groundwater at 2.6–3.9 m, seasonal variation expected	Site-wide	Sump pumping/well-pointing required for excavations below approximately 2.5 m; higher levels likely in spring
Boulder-rich horizon of unproven extent	Between BH-07 and BH-09, central-east of plot	Risk to piling and excavation productivity; further investigation recommended (Section 7a)
Unrecorded made ground, approx. 900 m ² , extent unproven	North-west corner	Not suitable for direct founding without remediation; possible watching-brief obligation (Section 7b)
Localised elevated sulphide indication, unconfirmed	Vicinity of BH-12, south-central area	Potential waste-classification and segregated-handling requirement pending test results (Section 7c)
Site falls approximately 4 m north to south	Site-wide	Influences cut/fill balance, surface drainage falls and the founding depth transition described above

9. Limitations of this Report

This report is based on ground conditions interpreted from fourteen discrete borehole locations and associated laboratory testing. Ground conditions, including the depth to rock, the presence and extent of boulders, made ground and the sulphide indication described in Section 7, may vary between and beyond the locations investigated, and cannot be guaranteed to be representative of the whole site. The recommendations in this report are preliminary and are intended to inform design development; they should be confirmed by inspection and, where noted, by further investigation, during detailed design and construction. This report does not constitute a contamination, ecological, archaeological or structural assessment, and does not address any contractual allocation of ground risk, which is a matter for the project's contract documents.

End of PDC-SS-001 Rev B